



EE2112 Electric Circuit and Lab

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Week 12 Practice Problems – Second Order Systems

- L34: 2nd Order Systems I
- L35: 2nd Order Systems II
- L36: 2nd Order Systems III (Simulation)

Submission Instructions:

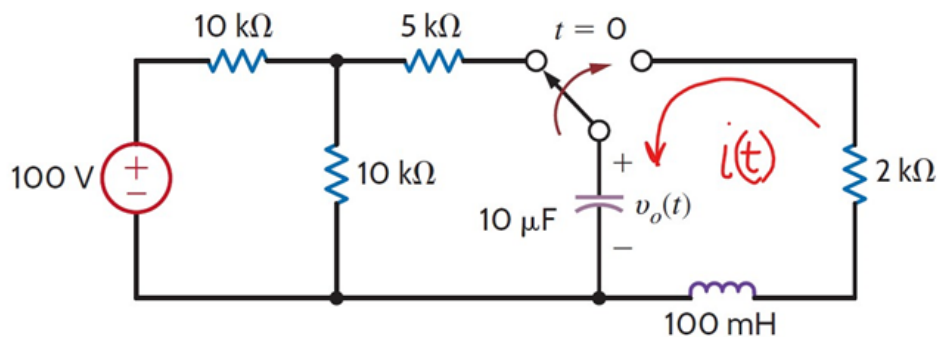
1. Submit your single file format (PDF or DOCX) via this Canvas assignment. This needs to be a separate file upload in the Canvas assignment.
 - a. This should include all the questions asked, in numerical order.
 - b. All pages should be in portrait view and easily readable.
 - c. This must include all of your work shown within this submission file. This includes but is not limited to, handwork, codes, plots/graphs.



1. Problem

1.1. Problem Statement

Using the differential equation approach, find an expression of the current $i(t)$ and voltage $v_o(t)$ for $t > 0$ in the circuit below.



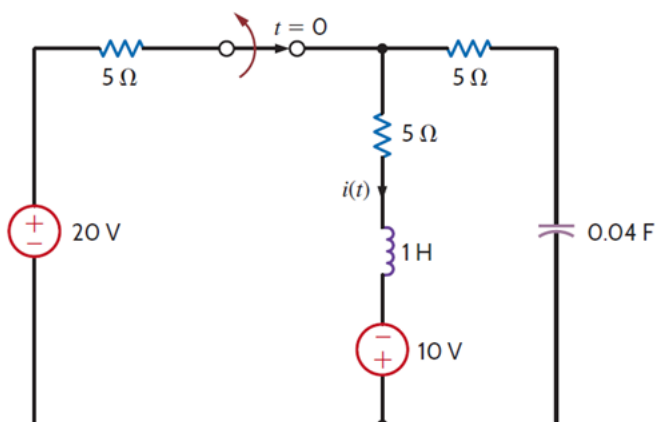
1.2. Solution



2. Problem

2.1. Problem Statement

Using the differential equation approach, find an expression of the current $i(t)$ for $t > 0$ in the circuit below.



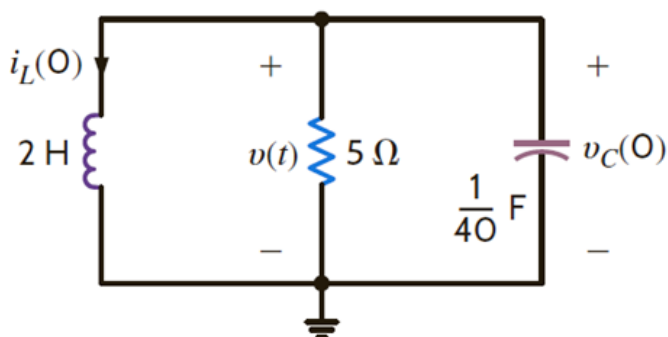
2.2. Solution



3. Problem

3.1. Problem Statement

Using the differential equation approach, find an expression of the voltage $v(t)$ for $t > 0$ in the circuit below. The initial conditions for the inductor and capacitor are $i_L(0) = 1\text{ A}$ and $v_C(0) = 10\text{ V}$ respectively.



3.2. Solution



4. Problem

4.1. Problem Statement

Design a parallel RLC circuit with a $R \geq 1k\Omega$ and has the characteristic equation shown below.

$$s^2 + 4 \times 10^7 s + 3 \times 10^{14} = 0 \quad (4.1)$$

4.2. Solution